

AMINE BENSAID

Software Engineer

Phone: +216 23051505 | Email: mohamedamine.bensaid@insat.ucar.tn

Location: Ariana, Tunisia | LinkedIn | GitHub

Summary

Software engineer with hands-on experience building distributed robotics systems, full-stack applications, and cloud-connected IoT solutions. ROS 2 developer on award-winning autonomous robotics projects, including 2nd place at EUROBOT 2026 Finals (France) and 1st place in NXP Cup Tunisia. Skilled in designing software across the full stack — from real-time embedded control and ROS 2 middleware to web dashboards, mobile applications, and cloud integration.

Experience

Software Engineer Intern (IoT) | Orange Digital Center | 4 months

- Engineered an end-to-end IoT solution for microalgae pool management, spanning embedded firmware, cloud integration, and full-stack monitoring.
- Developed embedded firmware for real-time sensor data acquisition and automated actuator control, implementing closed-loop logic for pool condition regulation.
- Integrated cloud infrastructure using AWS IoT Core and MQTT to stream live telemetry, enabling remote monitoring and command execution.
- Built a full-stack smart monitoring system with a web dashboard and touchscreen interface for real-time system visualization and control.
- Collaborated in an Agile team environment using Jira.

Projects

Eurobot 2026 - Autonomous Robotics System (ROS 2 Jazzy)

- Built an end-to-end autonomous robotics system using ROS 2 (Jazzy) for Eurobot, integrating autonomous navigation, task scheduling, Hybrid A* path planning, lidar-based perception, obstacle avoidance, and real-time robot control in a distributed multi-robot architecture.
- Implemented a micro-ROS communication layer between ROS 2 and embedded systems (STM32, ESP32) using Serial and UDP protocols for reliable low-latency real-time data exchange in heterogeneous embedded systems.
- Designed and integrated a robotics software stack on Raspberry Pi, combining ROS 2 nodes, micro-ROS agents, and embedded controllers (STM32/ESP32) for real-time autonomous execution in competition environments, including closed-loop actuator control using PID-based systems for precise motor regulation.
- Developed a Flask + ROSBridge full-stack robotics dashboard enabling live robot visualization, strategy control, simulation, system monitoring, and real-time telemetry over ROS topics.
- Deployed and maintained production robotics software on embedded Linux (Raspberry Pi) using systemd services for reliability and competition-grade system stability.

Lane-Following Robot (NXP Cup Project)

- Designed and implemented the software architecture for an autonomous lane-following robot using a Teensy 4 microcontroller, Pixy2 camera, and NXP components.
- Developed a real-time line-tracking system in Embedded C++, integrating PID control for precise motor speed adjustment based on visual lane input and encoder output.
- Contributed to the project's success in the NXP Cup, an international robotics competition held in the Netherlands (May 13-14), where the team was the only one qualified from Tunisia and achieved first place nationally.

4-Wheel Autonomous Line-Following Robot with PID Control System

- Developed an autonomous line-following robot with a 4-wheel drivetrain and servo-based steering, implementing a PID control system for real-time trajectory correction and stable line tracking.
- Designed and tuned PID algorithms for closed-loop control to improve tracking accuracy and minimize oscillations under varying surface and lighting conditions.
- Built a Flutter mobile application for real-time robot monitoring and control, enabling speed adjustment, remote command execution, and live telemetry visualization.
- Implemented Bluetooth Low Energy (BLE) communication between the robot and mobile application for low-latency data exchange.
- Integrated encoder-based odometry to estimate and visualize robot position and motion in real time within the mobile application.

Education

INSAT - Institut National des Sciences Appliquees et de Technologie

Engineer's Degree in Embedded Systems Engineering

Sep 2023 - Present

Skills

- Programming Languages: C/C++, Python, Dart, Java
- Web and Full-Stack: Flask, ROSBridge, REST APIs, WebSockets
- Mobile Development: Flutter
- Cloud and IoT: AWS IoT Core, MQTT, HTTP
- Robotics Software: ROS 2 (Jazzy), micro-ROS, Hybrid A* path planning, obstacle avoidance, odometry
- Embedded Platforms: STM32, ESP32, Teensy 4, Raspberry Pi
- Systems and Control: Real-time systems, firmware design, FreeRTOS, PID control
- Communication Protocols: I2C, SPI, UART, BLE, Serial, UDP
- Tools and Workflow: Git, GitHub, Jira, OpenCV, PlatformIO, systemd

Honors and Achievements

2nd Place — EUROBOT 2026 Finals — International Autonomous Robotics Competition, France (May 2026)

Represented Tunisia and INSAT at EUROBOT 2026 in France; achieved 2nd place in the Accumulation Phase (global) and 5th place overall in the Finals. This performance qualified the team through the international qualification rounds and placed us among the top teams worldwide.

1st Place - NXP Cup Tunisia and Finalist - International Robotics Competition (Netherlands, May 2025)

Represented Tunisia, INSAT, and AEROBOTIX INSAT as the only qualified Tunisian team in the international robotics competition held in the Netherlands. Achieved first place nationally with an autonomous lane-following robot, demonstrating advanced skills in embedded systems and real-time control.

1st Place - EUROBOT 2026 Qualifications

Won first place in the EUROBOT 2026 qualifications and qualified for the finals in France (May 2026), representing Tunisia among more than 100 teams from around the world.

Interests

- Hackathons and Coding Challenges: Solving real-world problems through team-based innovation and rapid prototyping.
- Tech and Sci-Fi Films: Inspired by futuristic concepts and AI-driven storytelling.